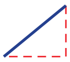
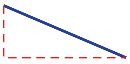
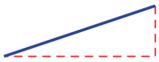
Exercise 6E

Understanding

1 Use the words ‘positive’, ‘negative’, ‘zero’ or ‘undefined’ to complete each sentence.
- a The gradient of a horizontal line is ______________.
- b The gradient of the line joining (0, 3) and (5, 0) is ____________.
- c The gradient of the line joining (-6, 0) and (1, 1) is _____________.
- d The gradient of a vertical line is ____________.

2 Decide whether each of the following lines would have a positive or negative gradient.

Lines going downhill from left to right have a negative gradient.

a 
b 
c 
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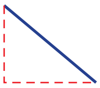
e 
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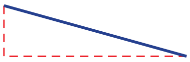
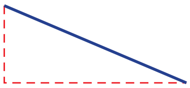
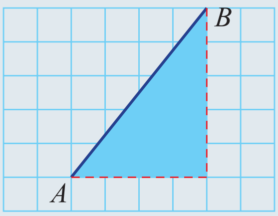
h 
i 

Example 14 Finding the gradient from a grid

Find the gradient of the following line segments, where each grid box equals 1 unit.

a


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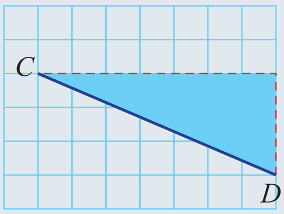
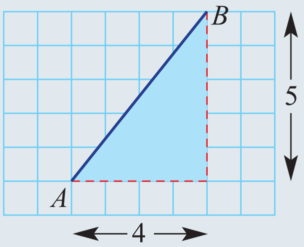


Solution | **Explanation**

a Gradient of AB $= \dfrac{\text{rise}}{\text{run}}$

$= \dfrac{5}{4}$

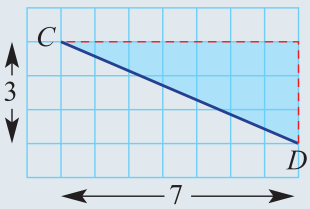


The slope is upwards and therefore the gradient is positive.
The rise is 5 and the run is 4.

b Gradient of CD $= \dfrac{\text{rise}}{\text{run}}$

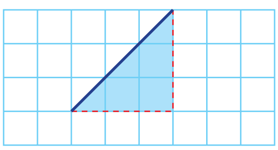
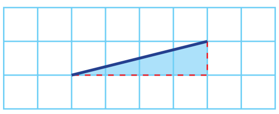
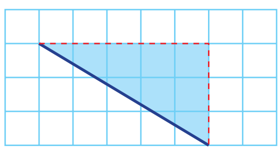
$= \dfrac{-3}{7}$

$= -\dfrac{3}{7}$



The slope is downwards and therefore the gradient is negative.
The fall is 3 so we write rise = -3, and the run is 7.

3 Find the gradient of the following line segments.

a 
b 
c 

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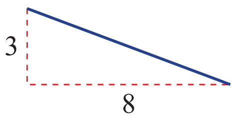
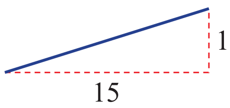
4 Find the gradient of the following.

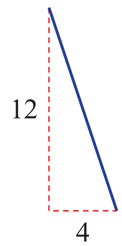
a



b



c



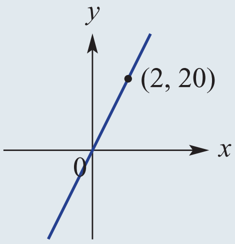
The gradient is written as a fraction or a whole number.

Fluency

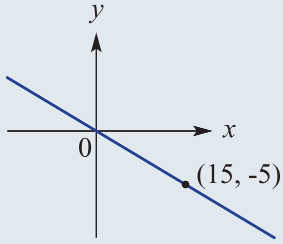
Example 15 Finding the gradient from graphs

Find the gradient of the following lines.

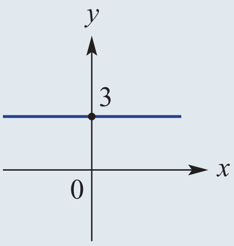
a



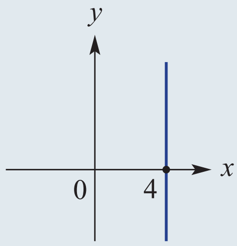
b



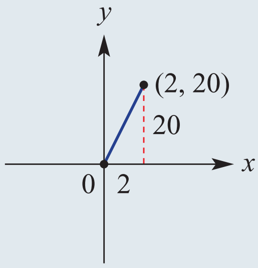
c



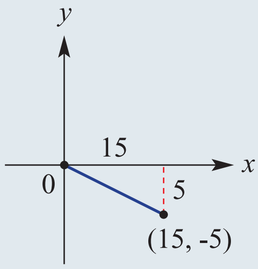
d



Solution	**Explanation**
a Gradient $= \dfrac{\text{rise}}{\text{run}}$	Write the rule each time.
$= \dfrac{20}{2}$	The rise is 20 and the run is 2 between the two points (0, 0) and (2, 20).
$= 10$	Simplify by cancelling.



b Gradient $= \dfrac{\text{rise}}{\text{run}}$	Note this time that, when working from left to right, there will be a slope downwards.
$= \dfrac{-5}{15}$	The fall is 5 (rise = -5) and the run is 15.
$= \dfrac{-1}{3}$	Simplify.
$= -\dfrac{1}{3}$	



c Gradient = 0	Horizontal lines have a zero gradient.
d Gradient is undefined.	Vertical lines have an undefined gradient.

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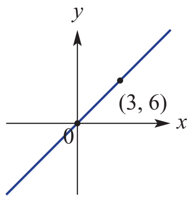
5 Find the gradient of the following lines.

a


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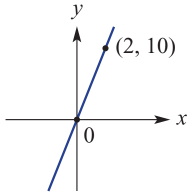

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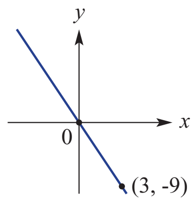

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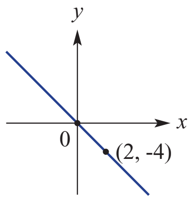

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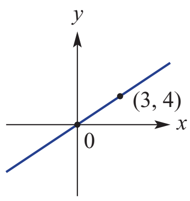

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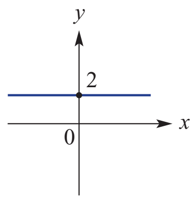

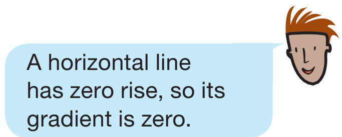

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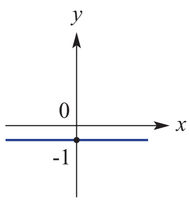

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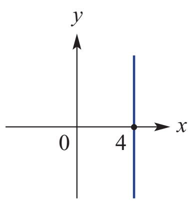

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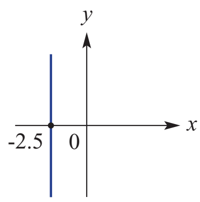
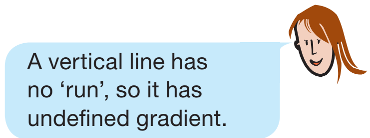
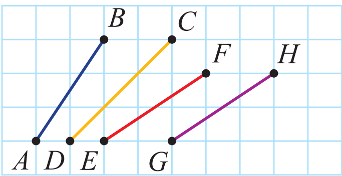
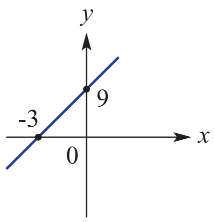




6 Use the grid to find the gradient of the following line segments. Then order the segments from least to greatest gradient.



7 Determine the gradient of the following lines.

a


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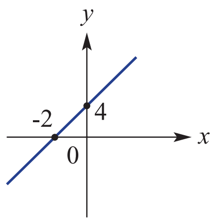

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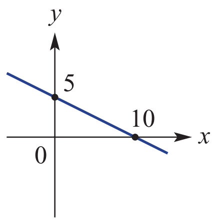

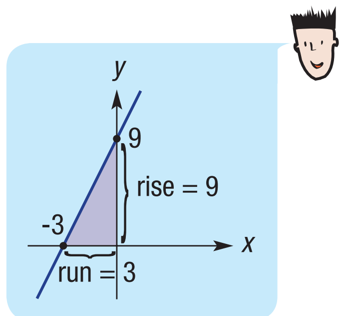

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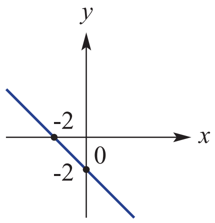

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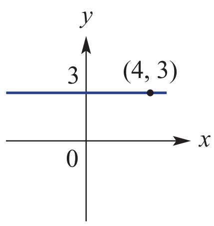

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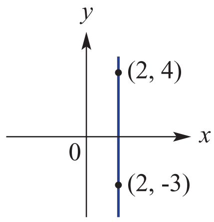

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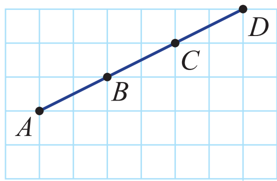
Problem-solving and Reasoning

8 a Copy and complete the table below.

Line segment	Rise	Run	Gradient
AB			
AC			
AD			
BC			
BD			
CD			

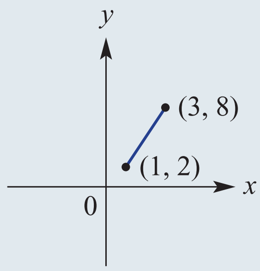


b What do you notice about the gradient between points on the same line?

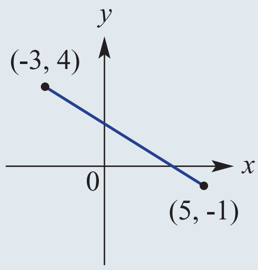
Example 16 Using a formula to calculate gradient

Use the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ to find the gradient of the line segments between the following pairs of points.

a



b



Solution	Explanation
a $m = \frac{y_2 - y_1}{x_2 - x_1}$ $= \frac{8-2}{3-1}$ $= \frac{6}{2}$ $= 3$	Write the rule. (1, 2) (3, 8) x_1 y_1 x_2 y_2 It does not matter which point is labelled (x_1, y_1) and which is (x_2, y_2).
b $m = \frac{y_2 - y_1}{x_2 - x_1}$ $= \frac{-1-4}{5-(-3)}$ $= \frac{-5}{8}$ $= -\frac{5}{8}$	Write the rule. (5, -1) (-3, 4) x_1 y_1 x_2 y_2 Remember that $5 - (-3) = 5 + 3 = 8$

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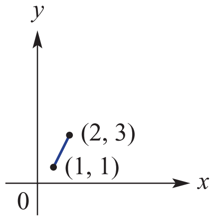
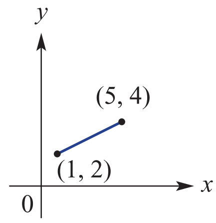
6E

9 Use the formula $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ to find the gradient between these pairs of points.

a



b



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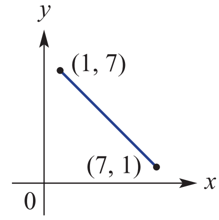
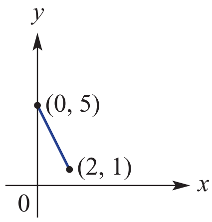


First copy the coordinates and label them.
E.g. (1, 1) (2, 3)
x_1 y_1 x_2 y_2

d



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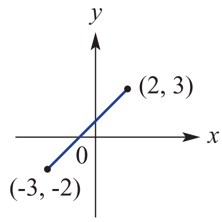

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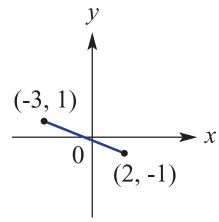


You can choose either point to be (x_1, y_1).

10 Find the gradient between the following pairs of points:

a (1, 3) and (5, 7) **b** (-1, -1) and (3, 3)
c (-3, 4) and (2, 1) **d** (-6, -1) and (3, -1)
e (1, -4) and (2, 7) **f** (-4, -2) and (-1, -1)

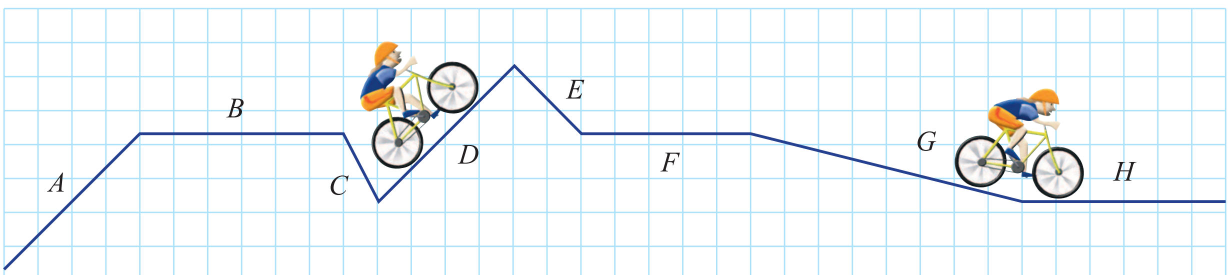
Use $m = \dfrac{y_2 - y_1}{x_2 - x_1}$.

11 The first section of the Cairns Skyrail travels from Caravonica terminal at 5 m above sea level to Red Peak terminal, which is 545 metres above sea level. This is across a horizontal distance of approximately 1.57 km. What is the overall gradient of this section of the Skyrail? Round the answer to three decimal places.

Both distances need to be in the same units.

From Bakersville to Rolland

12 A transversal map for a bike ride from Bakersville to Rolland is shown.



a Which sections A, B, C, D, E, F, G or H indicate travelling a positive gradient?
b Which sections indicate travelling a negative gradient?
c Which will be the hardest section to ride?
d Which sections show a zero gradient?
e Which section is the flattest of the downhill rides?

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